

## Python DA Assignment 2-Data Visualization

### 1) Loading the Taxis Dataset using the following code:

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Load the taxis dataset
df = sns.load_dataset("taxis")

df.head()
```

Matplotlib is building the font cache; this may take a moment.

|   | pickup                 | dropoff                | passengers | distance | fare | tip  | tolls | total | color  | payment        | pickup_zone              | dropoff_zone             | pickup_borough | dropoff_borough |
|---|------------------------|------------------------|------------|----------|------|------|-------|-------|--------|----------------|--------------------------|--------------------------|----------------|-----------------|
| 0 | 2019-03-23<br>20:21:09 | 2019-03-23<br>20:27:24 | 1          | 1.60     | 7.0  | 2.15 | 0.0   | 12.95 | yellow | credit<br>card | Lenox Hill West          | UN/Turtle Bay<br>South   | Manhattan      | Manhattan       |
| 1 | 2019-03-04<br>16:11:55 | 2019-03-04<br>16:19:00 | 1          | 0.79     | 5.0  | 0.00 | 0.0   | 9.30  | yellow | cash           | Upper West Side<br>South | Upper West Side<br>South | Manhattan      | Manhattan       |
| 2 | 2019-03-27<br>17:53:01 | 2019-03-27<br>18:00:25 | 1          | 1.37     | 7.5  | 2.36 | 0.0   | 14.16 | yellow | credit<br>card | Alphabet City            | West Village             | Manhattan      | Manhattan       |
| 3 | 2019-03-10<br>01:23:59 | 2019-03-10<br>01:49:51 | 1          | 7.70     | 27.0 | 6.15 | 0.0   | 36.95 | yellow | credit<br>card | Hudson Sq                | Yorkville West           | Manhattan      | Manhattan       |
| 4 | 2019-03-30<br>13:27:42 | 2019-03-30<br>13:37:14 | 3          | 2.16     | 9.0  | 1.10 | 0.0   | 13.40 | yellow | credit<br>card | Midtown East             | Yorkville West           | Manhattan      | Manhattan       |

### 2) Handling Missing Values

```
df.isnull().sum()
```

```
pickup          0
dropoff         0
passengers      0
distance        0
fare            0
tip             0
tolls           0
total           0
color           0
payment        44
pickup_zone     26
dropoff_zone    45
pickup_borough  26
dropoff_borough 45
dtype: int64
```

```
df.isnull().sum()[df.isnull().sum() > 0]
```

```
payment          44  
pickup_zone      26  
dropoff_zone     45  
pickup_borough   26  
dropoff_borough  45  
dtype: int64
```

```
df["payment"] = df["payment"].fillna(df["payment"].mode()[0])
```

```
df["payment"].isnull().sum()
```

```
np.int64(0)
```

```
df["pickup_borough"] = df["pickup_borough"].fillna(  
    df["pickup_borough"].mode()[0]  
)
```

```
df["pickup_borough"].isnull().sum()
```

```
np.int64(0)
```

```
df["dropoff_borough"] = df["dropoff_borough"].fillna(  
    df["dropoff_borough"].mode()[0]  
)
```

```
df["dropoff_borough"].isnull().sum()
```

```
np.int64(0)
```

```
df["pickup_zone"] = df["pickup_zone"].fillna(
    df["pickup_zone"].mode()[0]
)

df["pickup_zone"].isnull().sum()

np.int64(0)
```

```
df["dropoff_zone"] = df["dropoff_zone"].fillna(
    df["dropoff_zone"].mode()[0]
)

df["dropoff_zone"].isnull().sum()

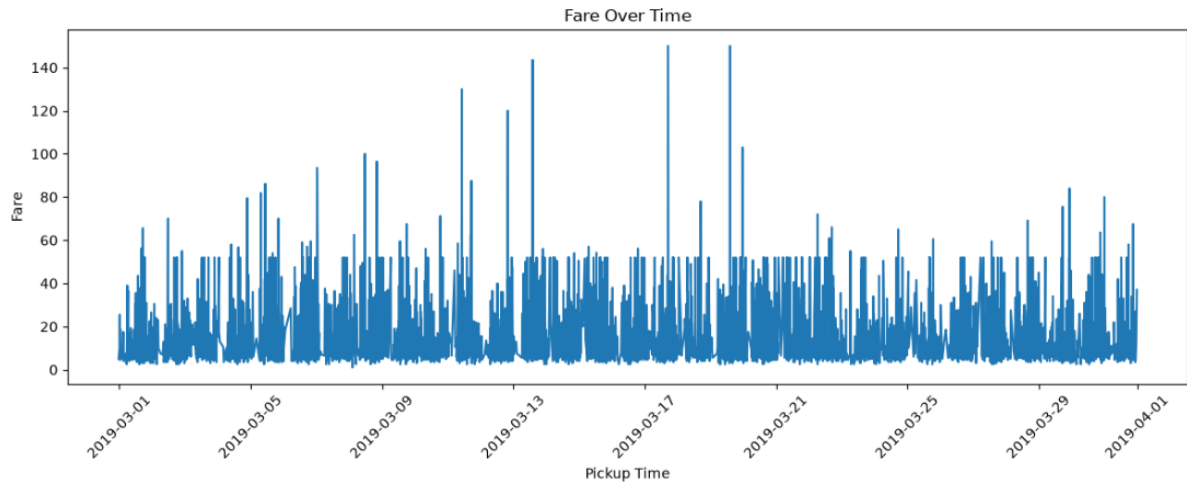
np.int64(0)
```

```
df.isnull().sum()
```

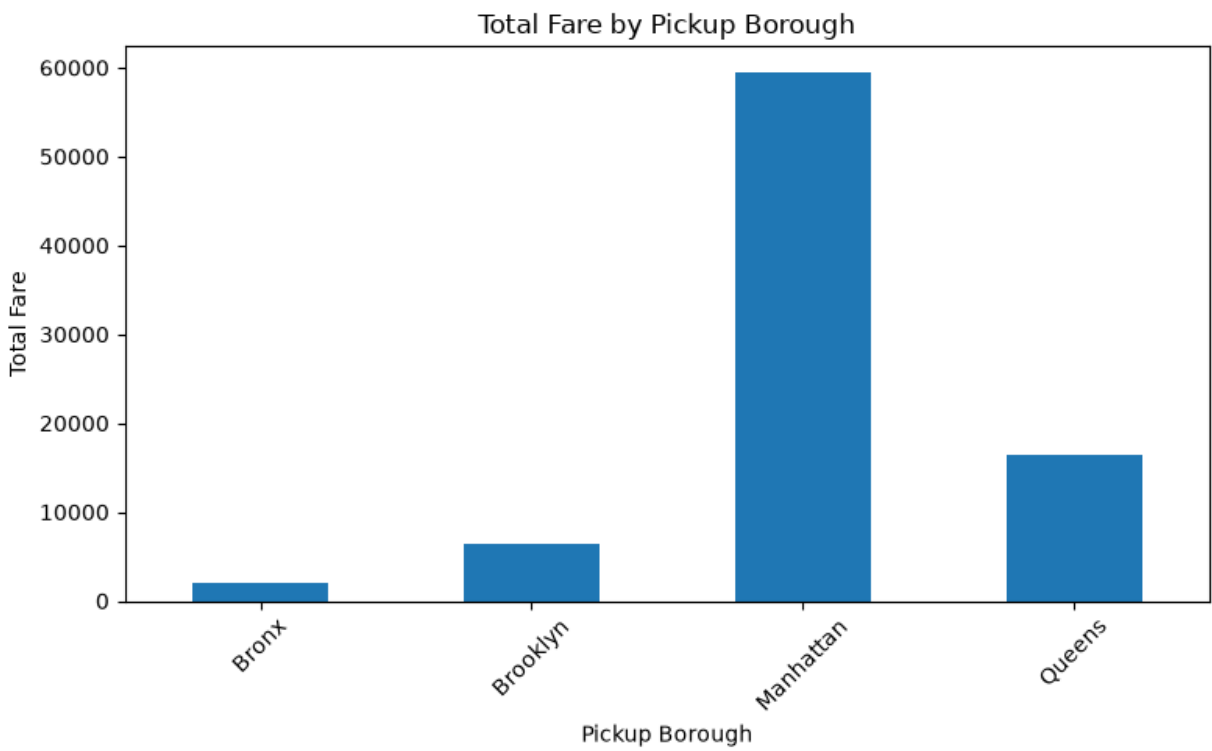
```
pickup          0
dropoff         0
passengers      0
distance        0
fare            0
tip             0
tolls           0
total           0
color           0
payment         0
pickup_zone     0
dropoff_zone    0
pickup_borough  0
dropoff_borough 0
dtype: int64
```

### 3) Visualizations using Matplotlib/Pandas Plot:

#### ★ Line Chart

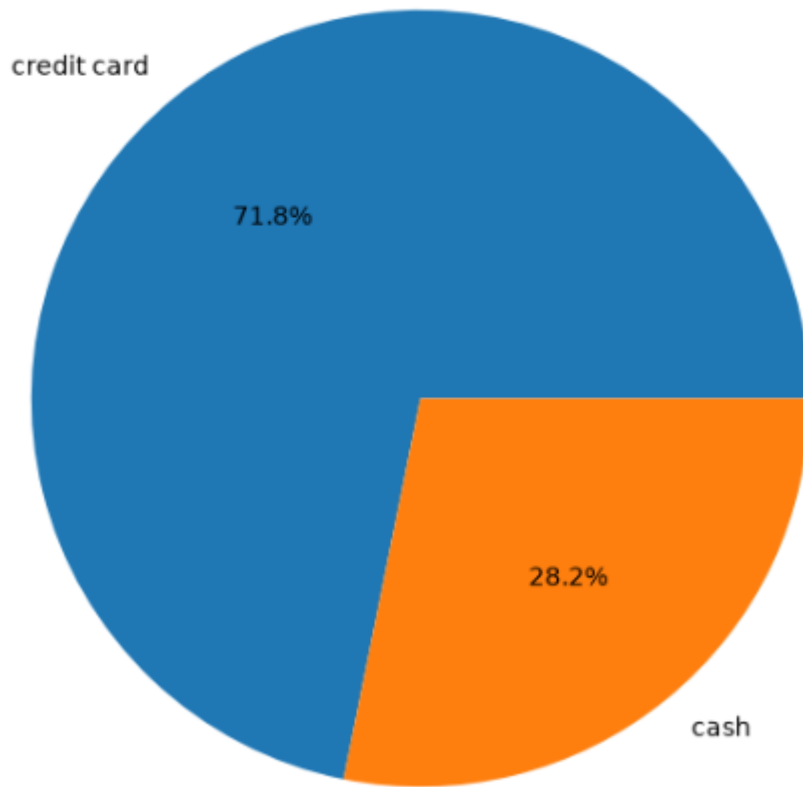


★ Bar Chart

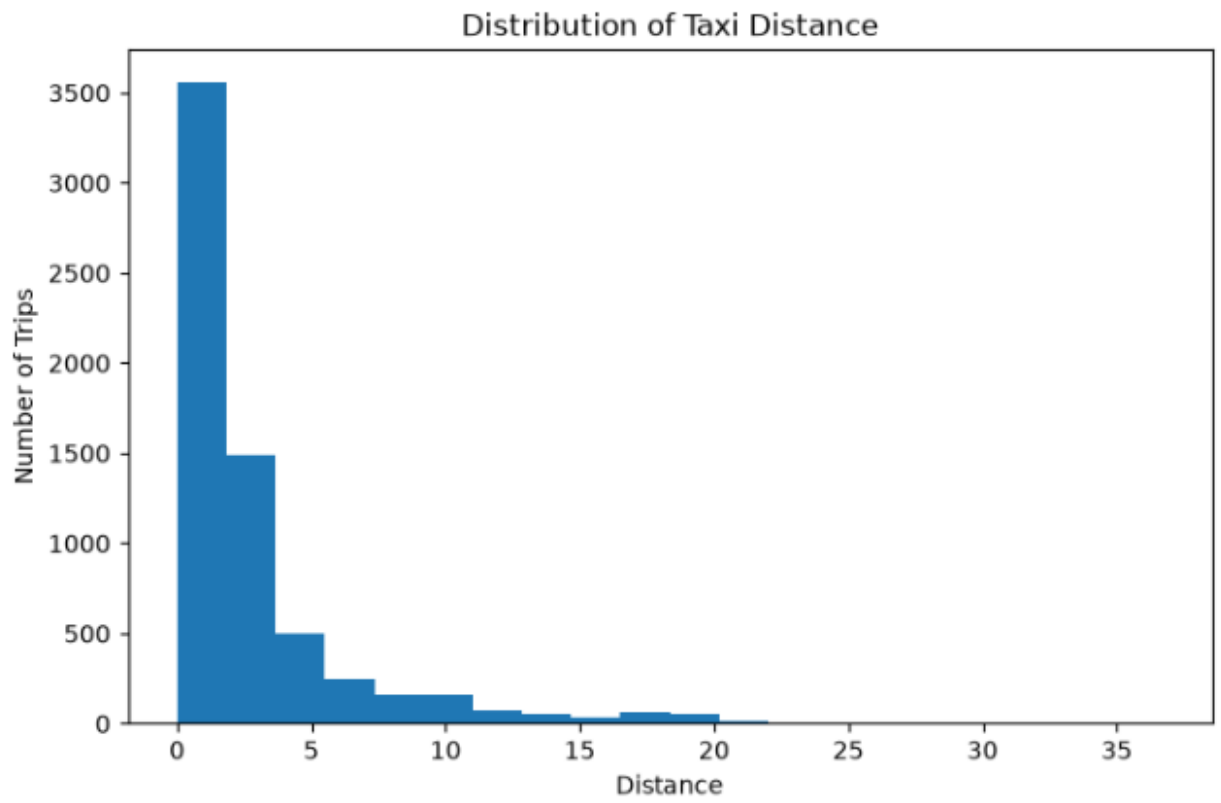


★ Pie Chart

Trip Distribution by Payment Method



★ Histogram



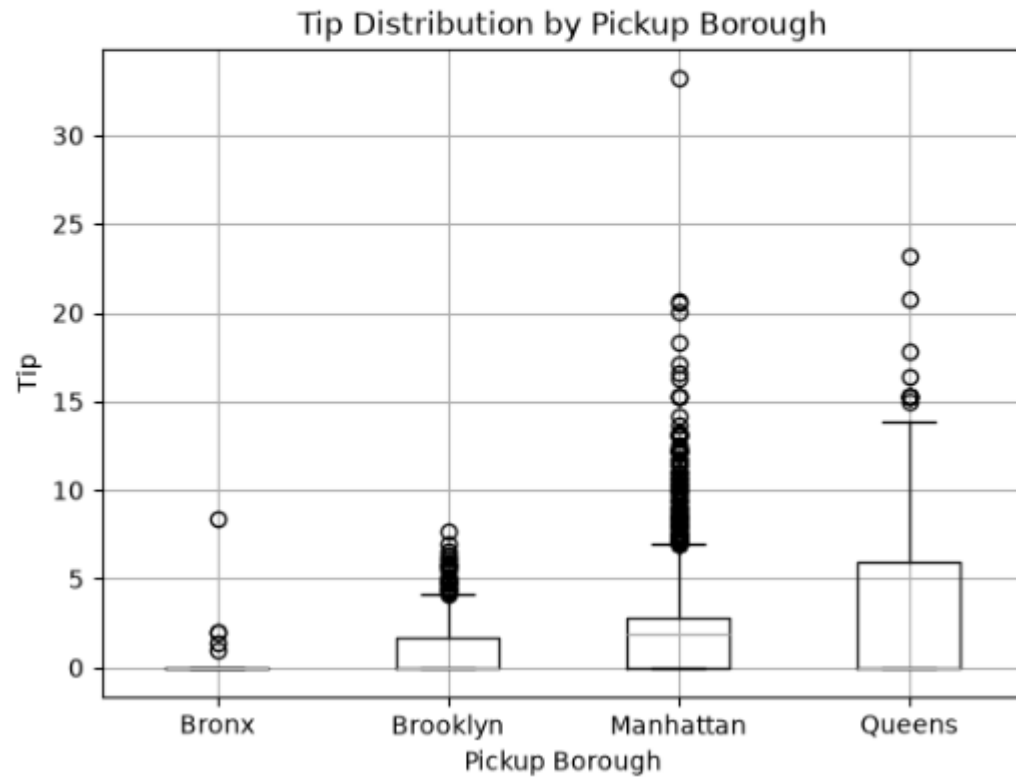
★ Box Plot

```
plt.ylabel("Tip")
plt.title("Tip Distribution by Pickup Borough")

plt.suptitle("")

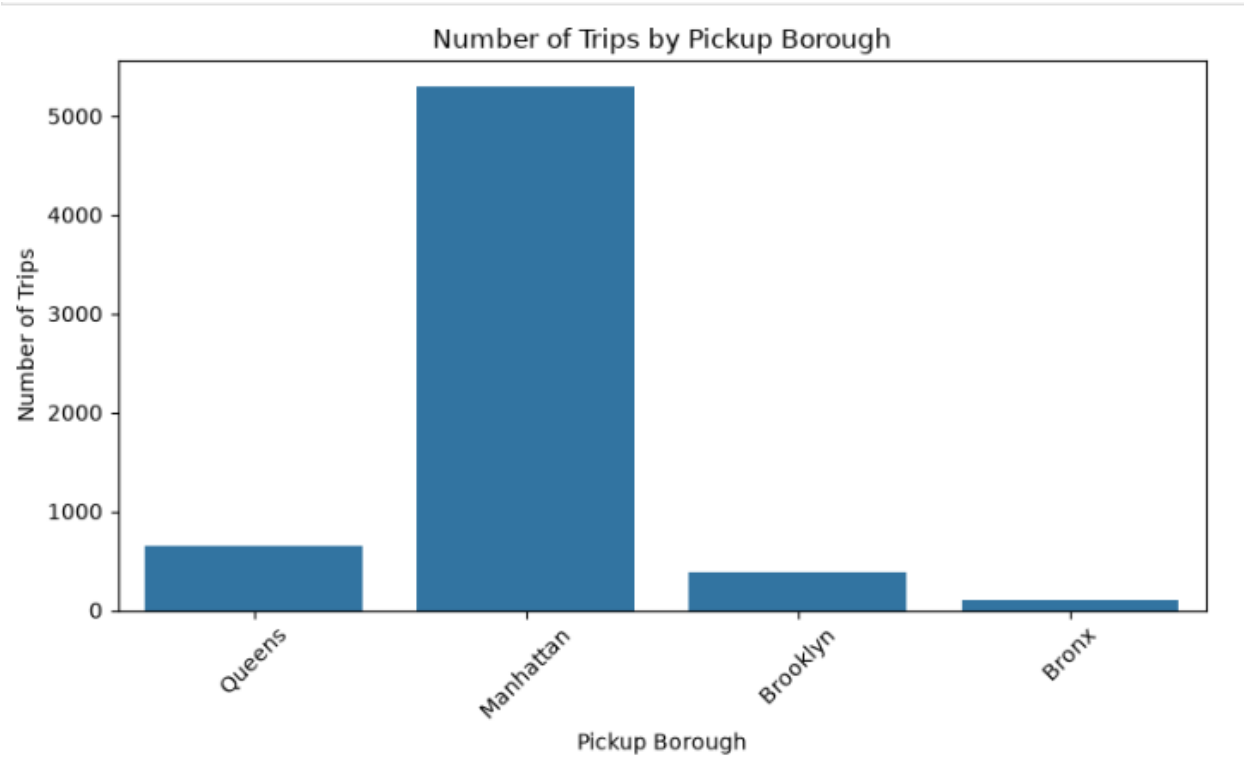
plt.show()
```

<Figure size 1000x500 with 0 Axes>



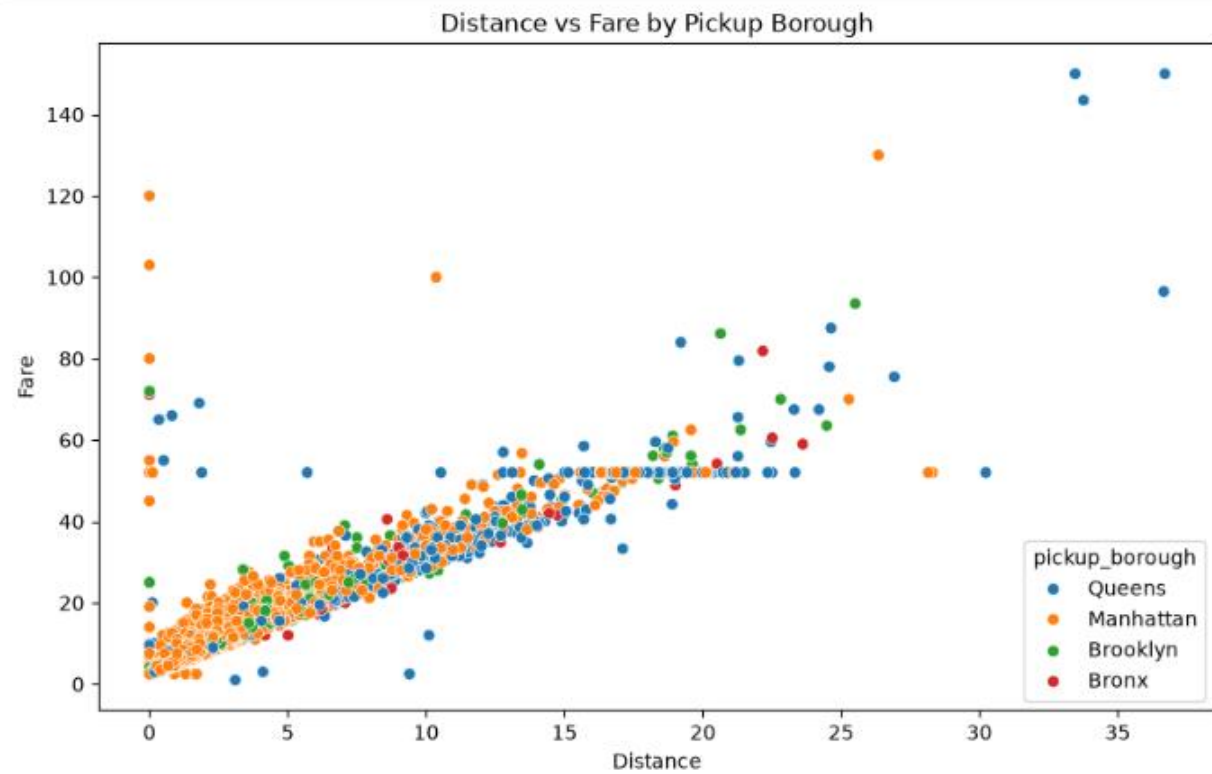
Visualizations using Seaborn:

★ Count Plot

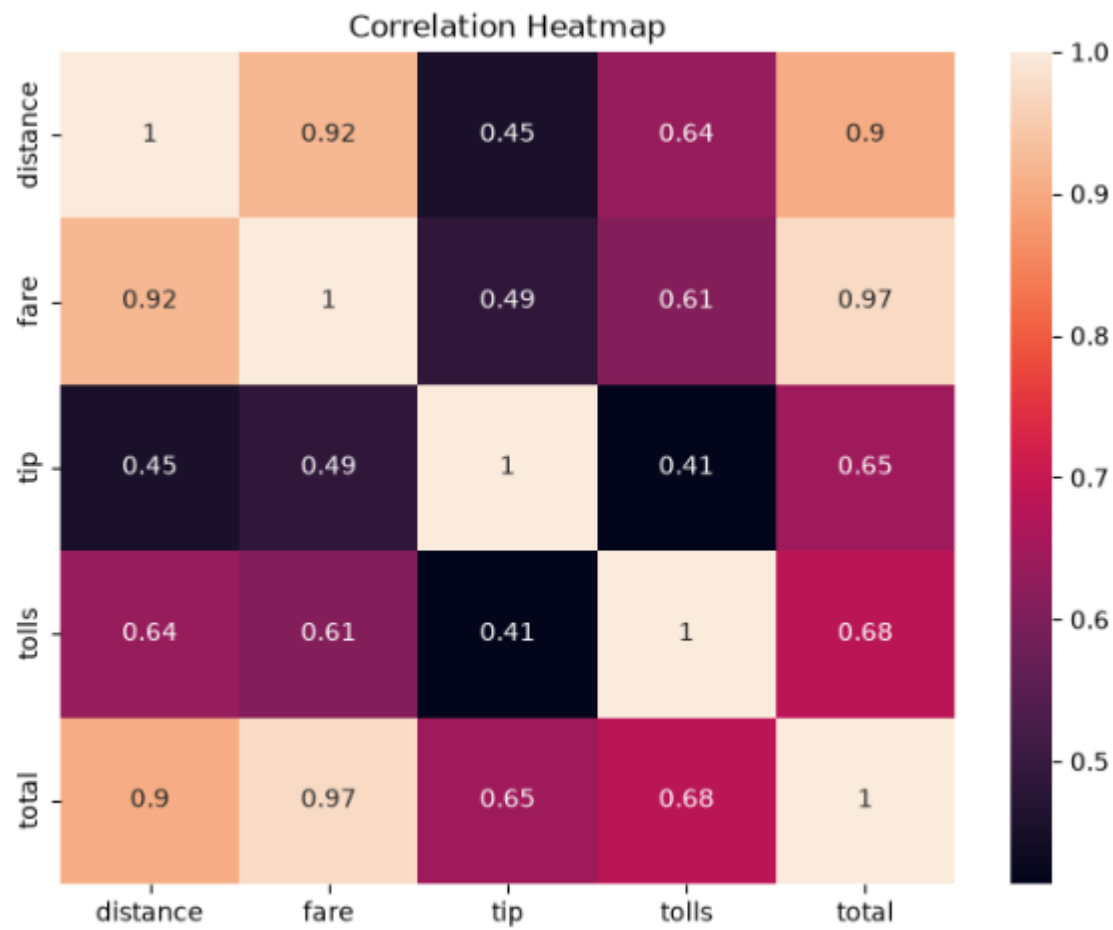


★ Scatter Plot

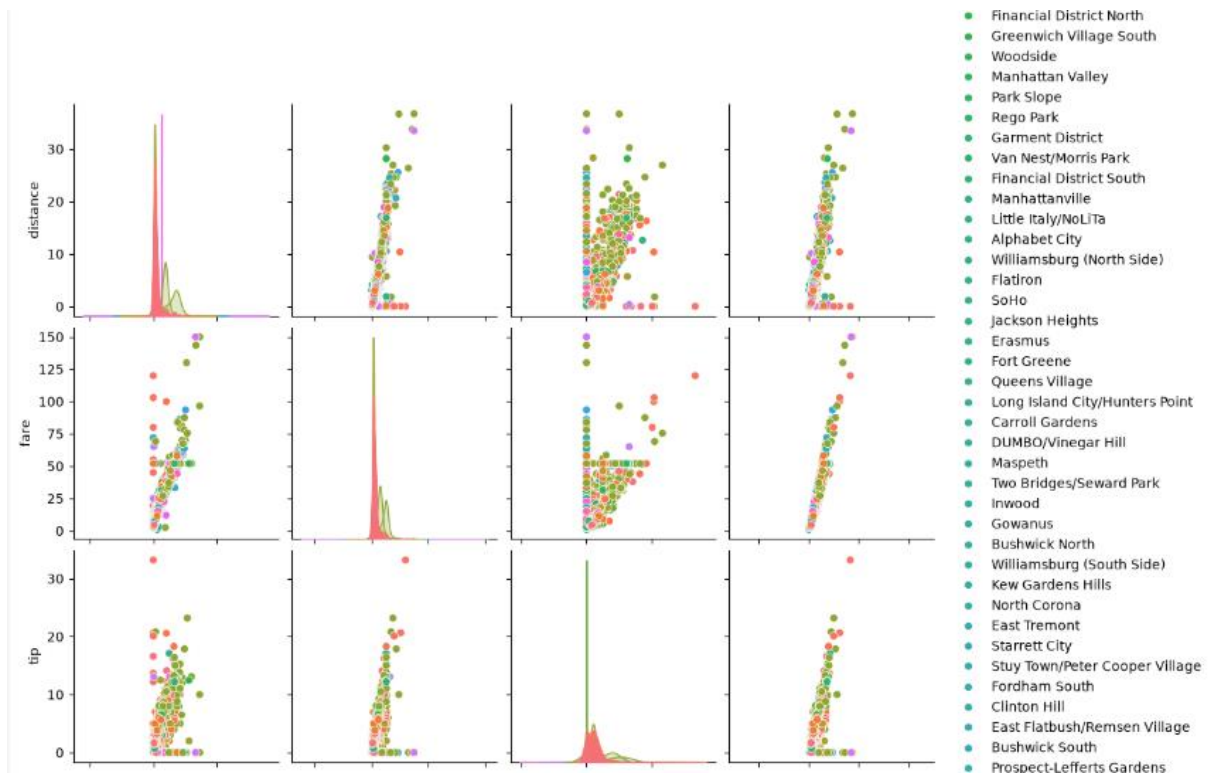
```
plt.ylabel("Fare")
plt.title("Distance vs Fare by Pickup Borough")
plt.show()
```



★ Heatmap



★ Pair Plot



★ Violin Plot

